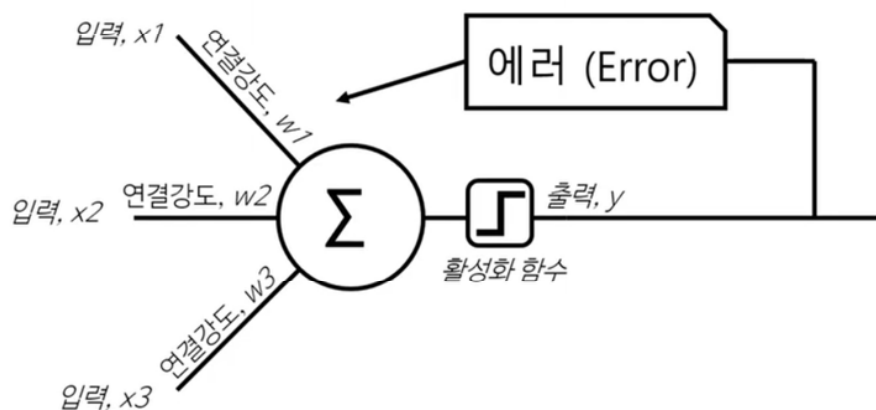


- [Neural Network 9] 역전파 backpropagation 알고리즘 (수정본)
 - https://youtu.be/DMCJ_GjBXwc?si=twlRQCe3G5PTnX5W
 - 출처 : 신박Ai 유튜브
 - 원본 출처 (EZlearn AI) :
<https://www.youtube.com/watch?v=QfIXxNfMCKo>
- **The Most Important Algorithm in Machine Learning**
 - 출처 : Artem Kirsanov 유튜브
 - <https://www.youtube.com/watch?v=SmZmBKc7Lrs>

564

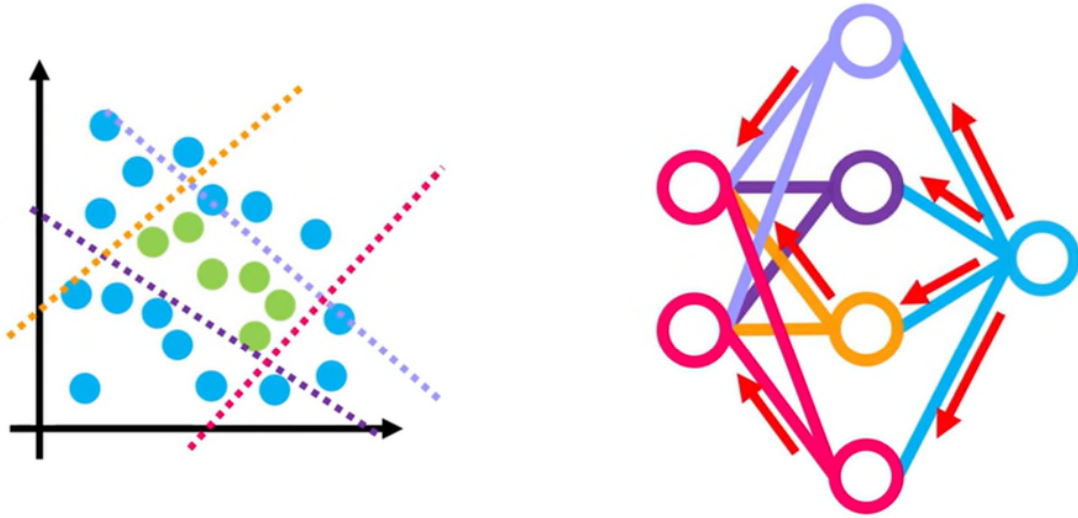
역전파는 실제로 무엇을 하는가? | 3장, 딥러닝 - 동영상 요약

단일 퍼셉트론(Perceptron)과 같은 얇은 신경망에 초점을 맞추고 있습니다



565

역전파 알고리즘이 개발되기 전까지는 해결하기 어려운 문제였습니다



566

Ronald J. Williams 등의 연구진이 역전파 알고리즘을 발표하였습니다



<https://www.psychologicalscience.org/observer/david-rumelhart>

David Rumelhart
수리심리학자



<https://www.frontiersofknowledgeawards-fbbva.es/galardonados/geoffrey-hinton-2/>

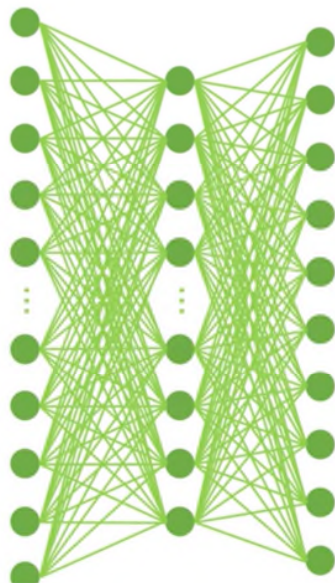
Geoffrey Hinton
실험심리학자
컴퓨터과학자



<https://www.ccs.neu.edu/home/rjw/>

Ronald J. Williams
컴퓨터과학자

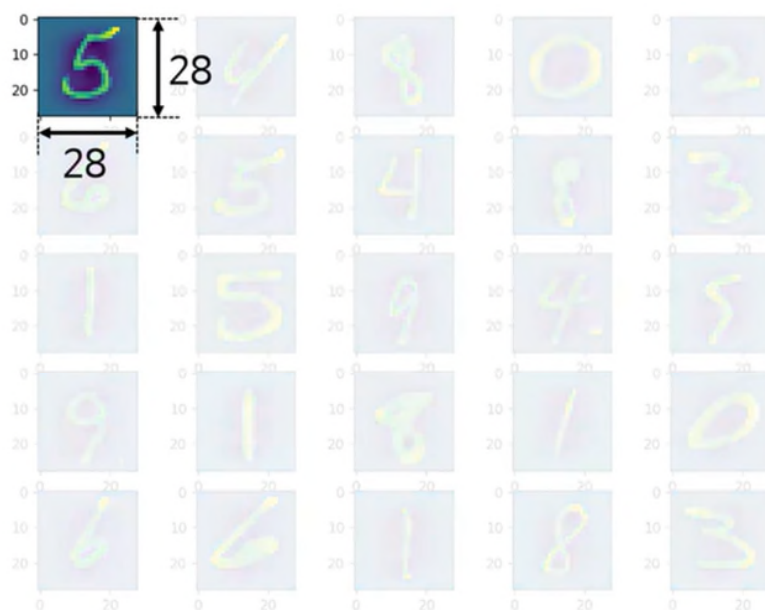
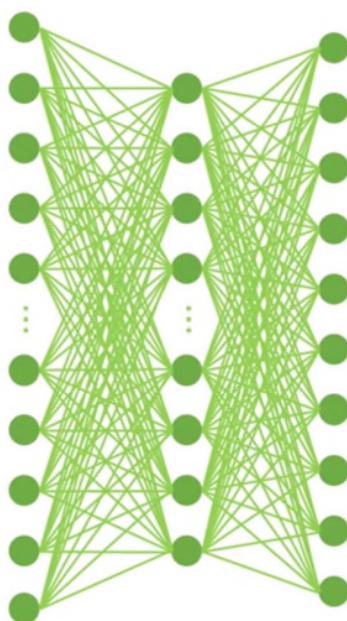
567



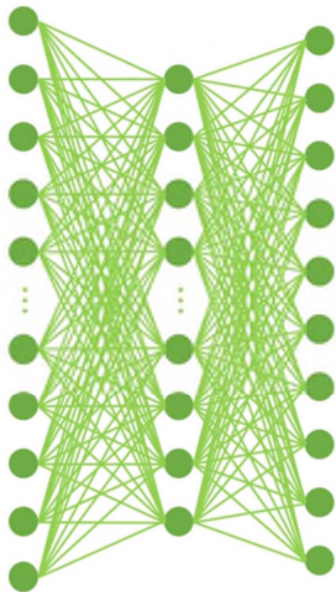
MNIST database. (2023, May 24). In *Wikipedia*.
https://en.wikipedia.org/wiki/MNIST_database

568

한 이미지는 784개의 데이터포인트 (=28x28)로 구성되어 있습니다

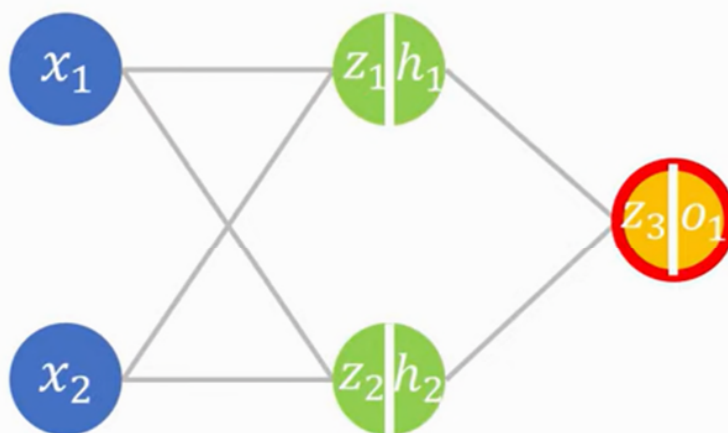


569

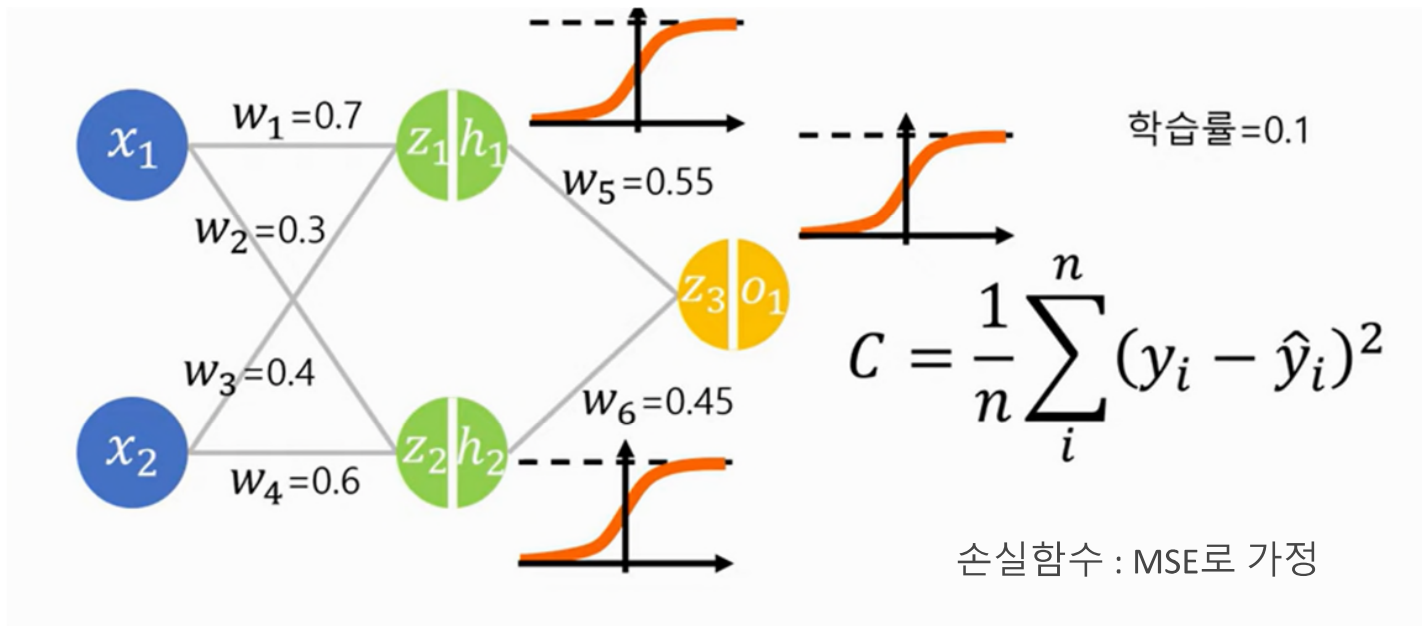


- 연결가중치 개수:
 $784 \times 100 + 100 \times 10 = 79,400$
- 편향 개수:
 $100 + 10 = 110$
- 한 개의 이미지 손실계산:
 $79,400$ 번의 연산량 (가중치만 고려)
- 한 개의 파라미터를 업데이트 하기 위해서는:
 $4,764,000,000 (=79,400 \times 60,000)$
- 모든 파라미터들에 대해 각각 연산하여야 하므로
 $4,767,000,000 \times 79,511 (=79,400 + 110 + 1) = 378,790,404,000,000$
- $378,790,404,000,000 / 850,000,000 = 445,635\text{초} = 123\text{시간}$

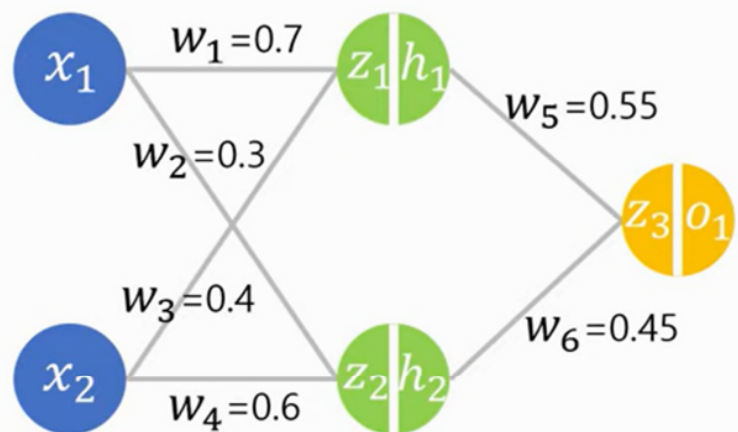
570



571



572



1단계: feedforward 순전파

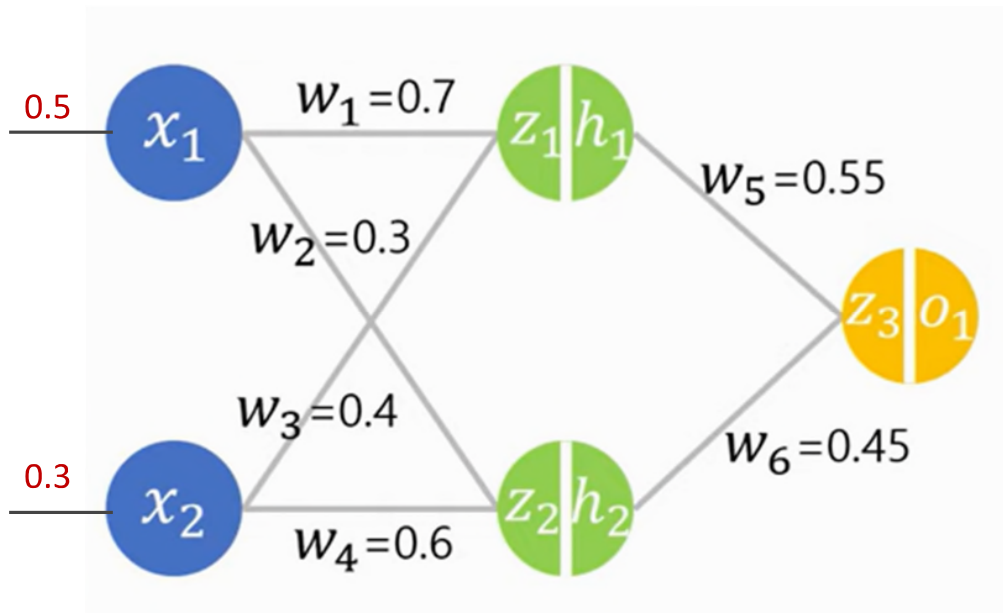
2단계: 손실계산

3단계: backpropagation 역전파

학습 (최적화 과정)

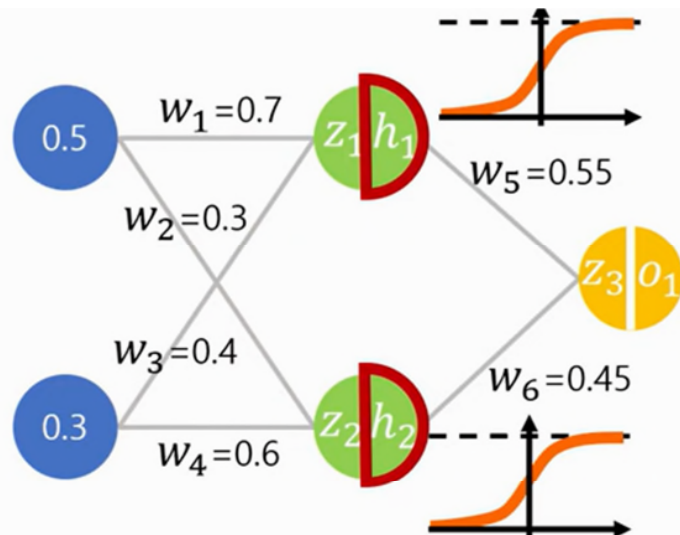
573

1단계: feedforward 순전파는 간단합니다



574

역전파는 실제로 무엇을 하는가? | 3장, 딥러닝 - 동영상 요약



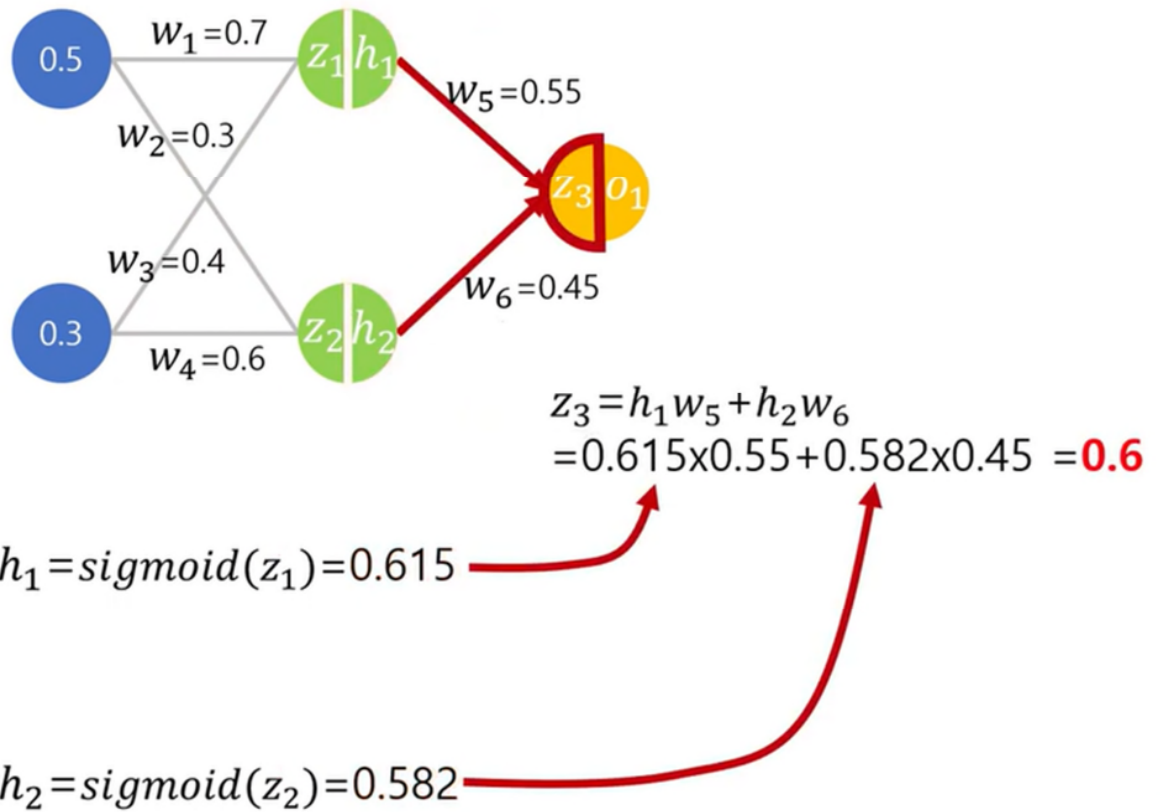
$$z_1 = x_1 w_1 + x_2 w_3 = 0.5 \times 0.7 + 0.3 \times 0.4 = 0.47$$

$$h_1 = \text{sigmoid}(z_1) = \mathbf{0.615}$$

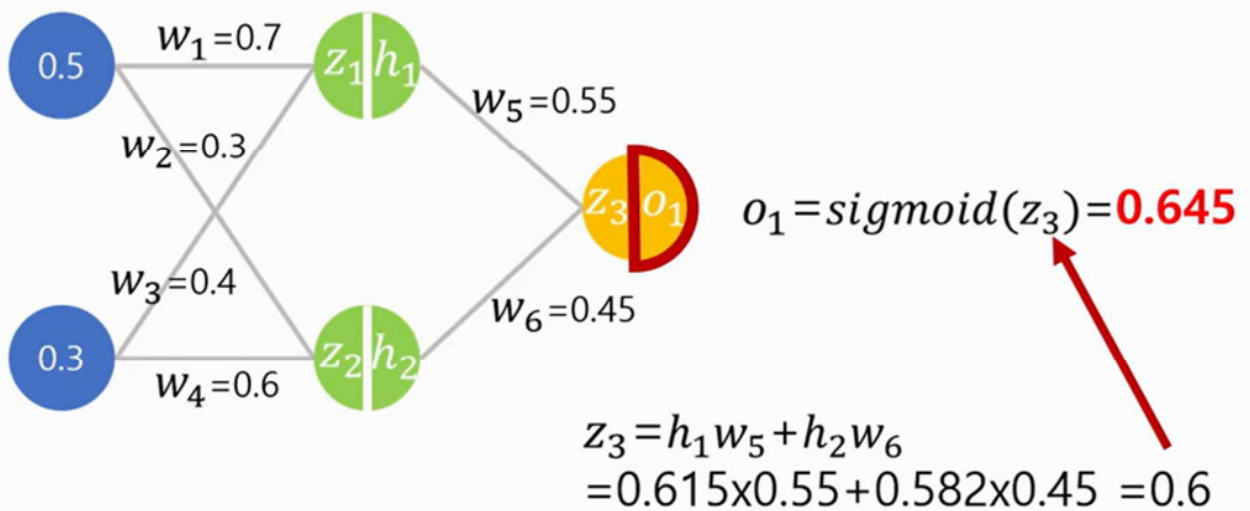
$$z_2 = x_1 w_2 + x_2 w_4 = 0.5 \times 0.3 + 0.3 \times 0.6 = 0.33$$

$$h_2 = \text{sigmoid}(z_2) = \mathbf{0.582}$$

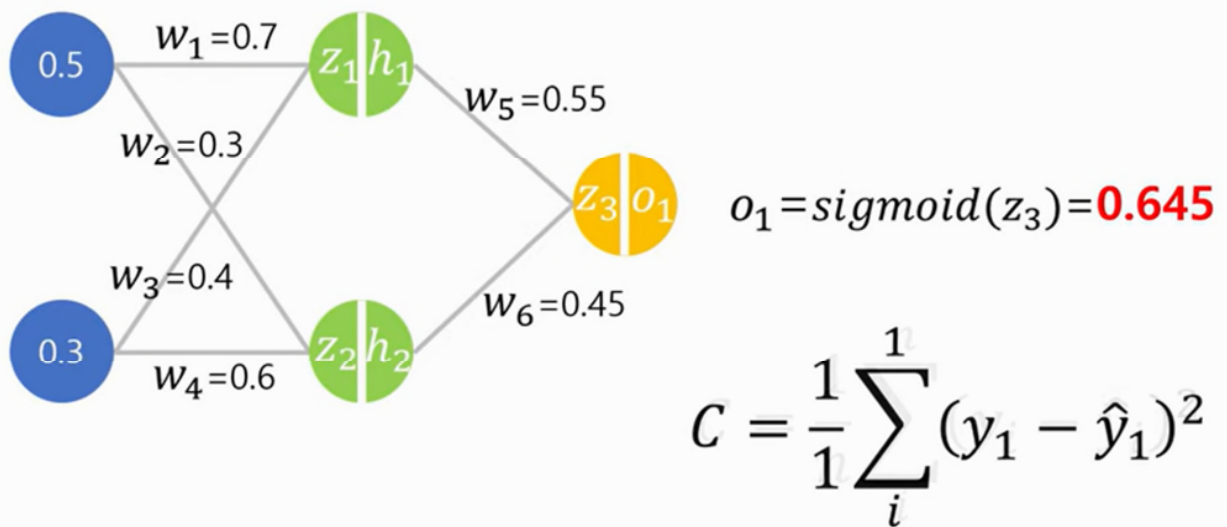
575



576



577

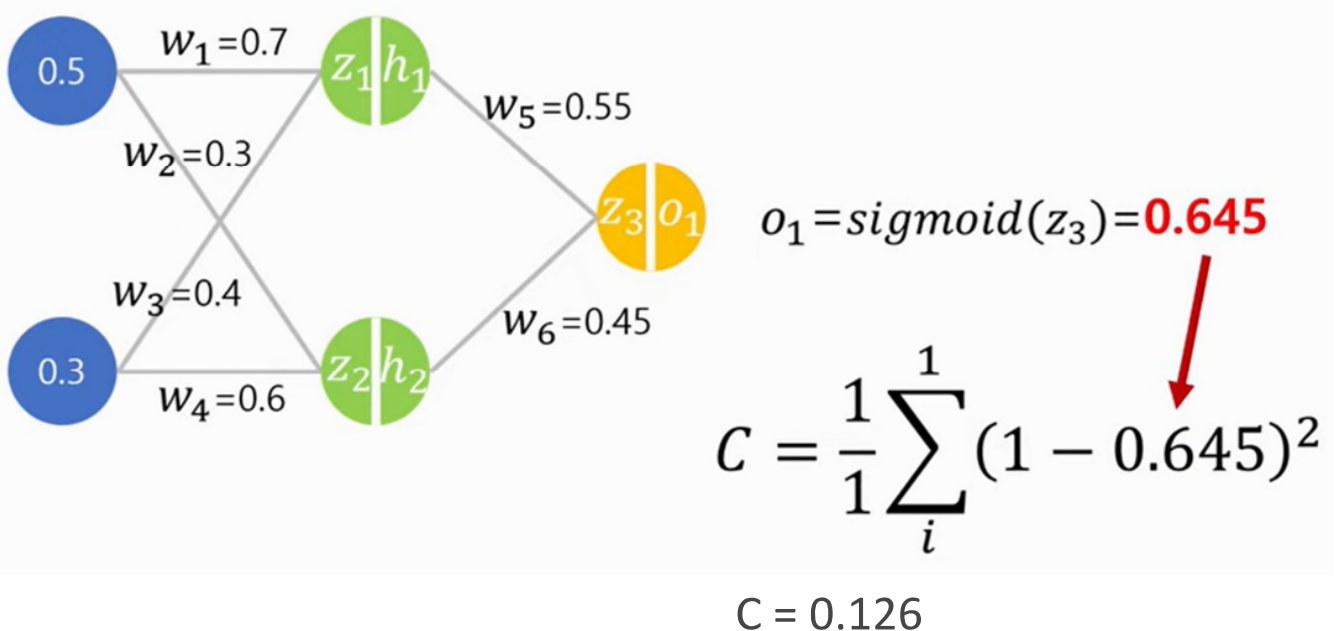


손실 : MSE를 사용할 경우 가정

실제 : binary cross entropy 사용

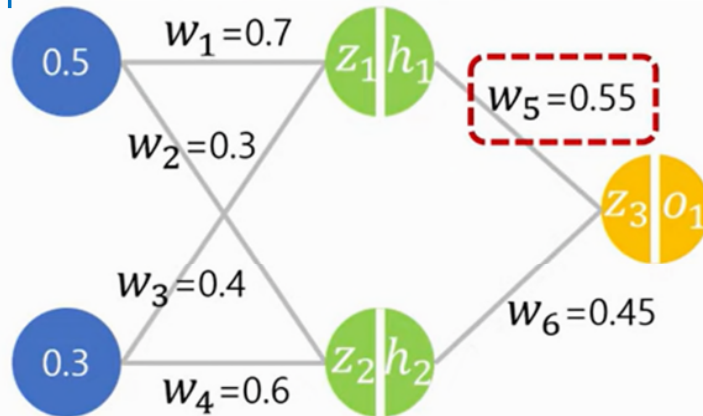
578

실제값을 1이라고 가정하면,



579

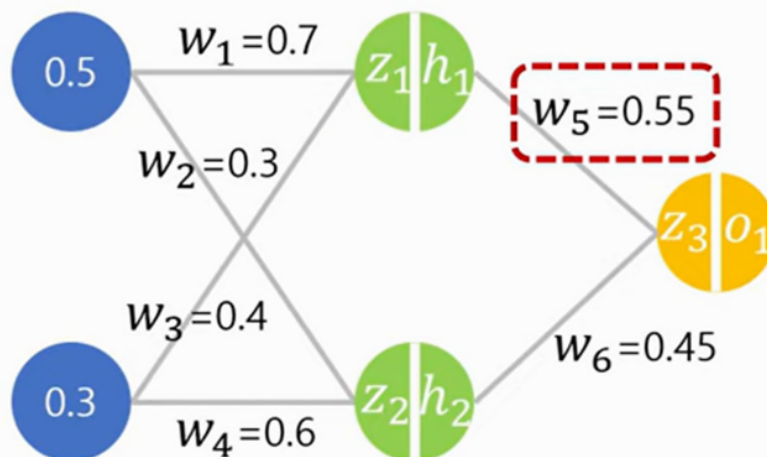
역전파 시작



경사하강법을 이용한 가중치 학습법:

$$\text{새 연결강도} = \text{현 연결강도} + \text{현 입력값} \times \frac{\partial C}{\partial w_5} \times \text{학습률}$$

580



$$\frac{\partial C}{\partial w_5} = \frac{\partial C}{\partial o_1} \cdot \frac{\partial o_1}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_5}$$

Chain rule

$$\frac{\partial C}{\partial w_5} = \frac{\partial C}{\partial o_1} \cdot \frac{\partial o_1}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_5}$$

581

●체인룰

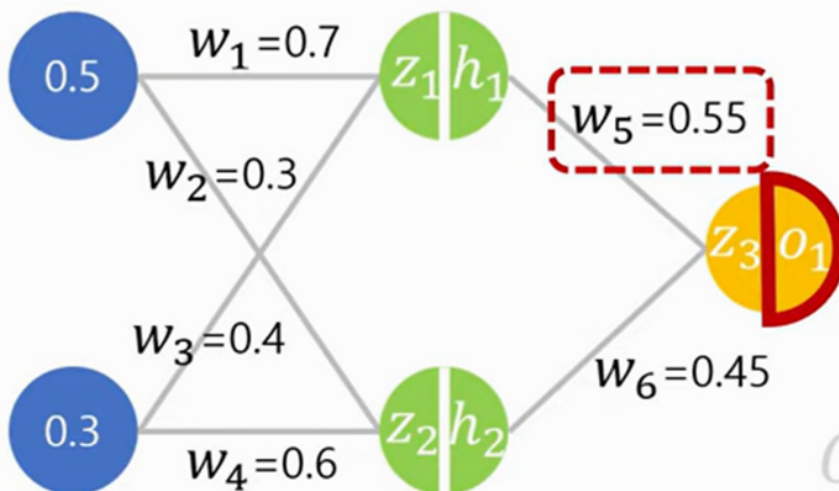
$$\frac{d\text{치타}}{d\text{사람}} = \frac{d\text{치타}}{d\text{사자}} \cdot \frac{d\text{사자}}{d\text{곰}} \cdot \frac{d\text{곰}}{d\text{사람}}$$

$$6 = 2 \times 2 \times 1.5$$

치타는 사자보다 2배 빠르고
 사자는 곰보다 2배 빠르고
 곰은 사람보다 1.5배 빠르다고 할때,
 치타는 사람보다 몇배 빠르나?
 직관적으로 $2 \times 2 \times 1.5 = 6$ 배

582

역전파는 실제로 무엇을 하는가? | 3장, 딥러닝 - 동영상 요약



$$o_1 = \text{sigmoid}(z_3) = \mathbf{0.645}$$

$$C = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$$

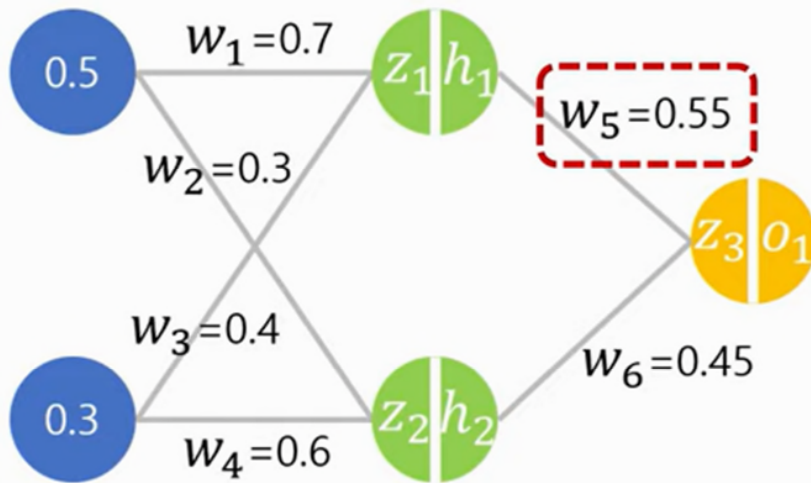
$$C = (y - o_1)^2$$

$$\frac{\partial C}{\partial o_1} = -2(y - o_1)^{2-1}$$

$$\begin{aligned} \frac{\partial C}{\partial o_1} &= -2(1 - 0.645) \\ &= -0.71 \end{aligned}$$

$$\frac{\partial C}{\partial w_5} = \frac{\partial C}{\partial o_1} \cdot \frac{\partial o_1}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_5}$$

583



$$o_1 = \text{sigmoid}(z_3) = \mathbf{0.645}$$

$$\frac{\partial C}{\partial w_5} = -0.71 \cdot \frac{\partial o_1}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_5}$$

584

$$S(x) = \frac{1}{1 + e^{-x}}$$

$$o_1 = \text{sigmoid}(z_3) = \mathbf{0.645}$$

$$O(z) = \frac{1}{1 + e^{-z}}$$

$$\frac{\partial O}{\partial z} = \frac{\partial}{\partial z} \left(\frac{1}{1 + e^{-z}} \right) = O(z)(1 - O(z))$$

585

$$S(x) = \frac{1}{1 + e^{-x}}$$

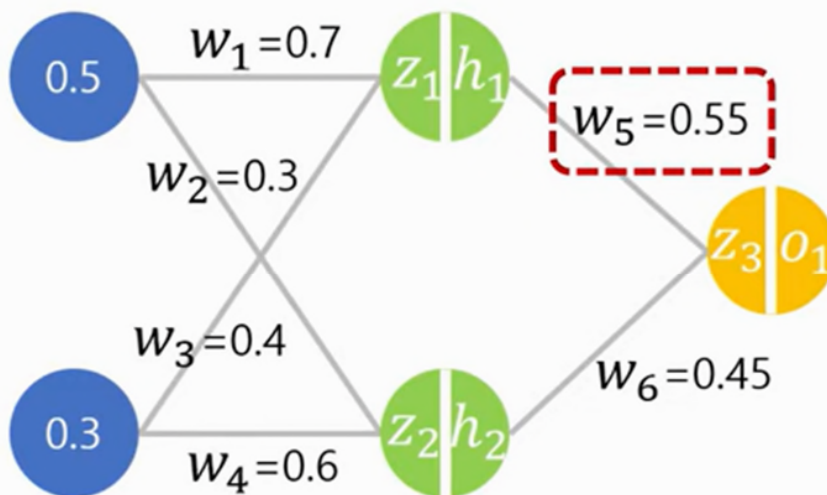
$$O(z) = \frac{1}{1 + e^{-z}}$$

$$\frac{\partial O}{\partial z} = \frac{\partial}{\partial z} \left(\frac{1}{1 + e^{-z}} \right) = O(z)(1 - O(z)) = 0.229$$

$$o_1 = \text{sigmoid}(z_3) = 0.645$$

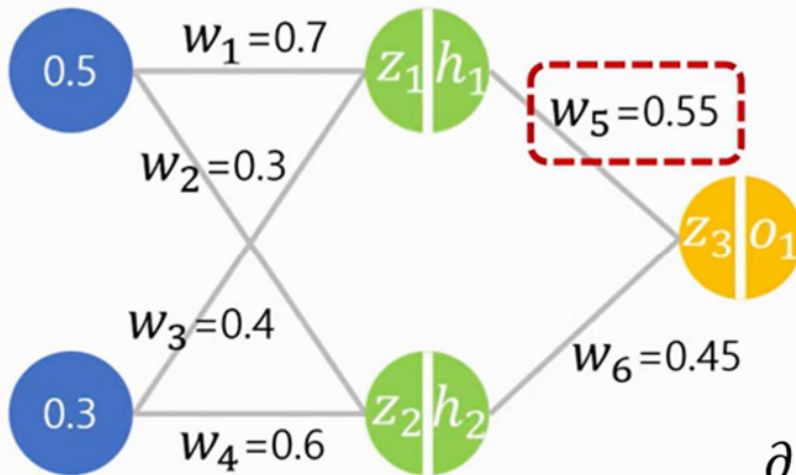
$$\frac{\partial O}{\partial z} = \frac{\partial}{\partial z} \left(\frac{1}{1 + e^{-z}} \right) = 0.645(1 - 0.645) = 0.229$$

586



$$\frac{\partial C}{\partial w_5} = -0.71 \cdot 0.229 \cdot \frac{\partial z_3}{\partial w_5}$$

587



$$z_3 = h_1 w_5 + h_2 w_6$$

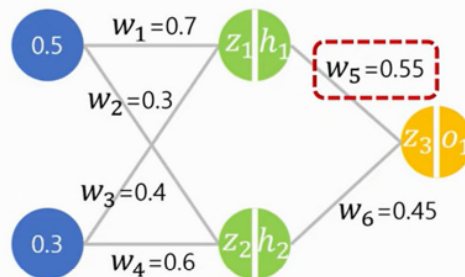
$$\frac{\partial z_3}{\partial w_5} = h_1 \cancel{w_5} + \cancel{h_2 w_6}$$

$$= h_1 = 0.615$$

(순전파에서 계산한 값)

$$\frac{\partial C}{\partial w_5} = -0.71 \cdot 0.229 \cdot \frac{\partial z_3}{\partial w_5}$$

588



$$z_3 = h_1 w_5 + h_2 w_6$$

$$\frac{\partial z_3}{\partial w_5} = h_1 \cancel{w_5} + \cancel{h_2 w_6}$$

$$= h_1 = 0.615$$

$$\frac{\partial C}{\partial w_5} = -0.71 \cdot 0.229 \cdot \frac{\partial z_3}{\partial w_5}$$

$$\frac{\partial C}{\partial w_5} = -0.71 \cdot 0.229 \cdot 0.615 = -0.01$$

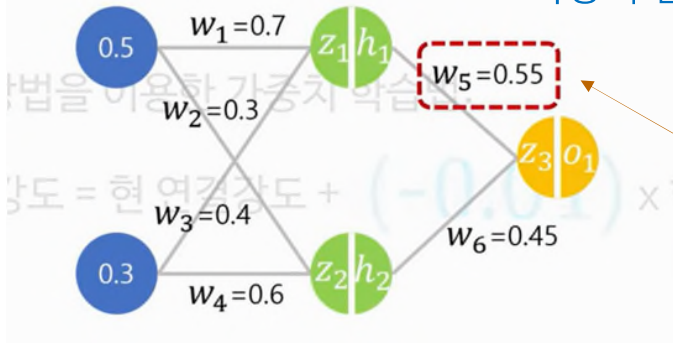
589

경사하강법을 이용한 가중치 학습법:

새 연결강도 = 현 연결강도 + $\frac{\partial C}{\partial w_5}$ x 학습률

$$\frac{\partial C}{\partial w_5} = -0.71 \cdot 0.229 \cdot 0.615 = -0.01$$

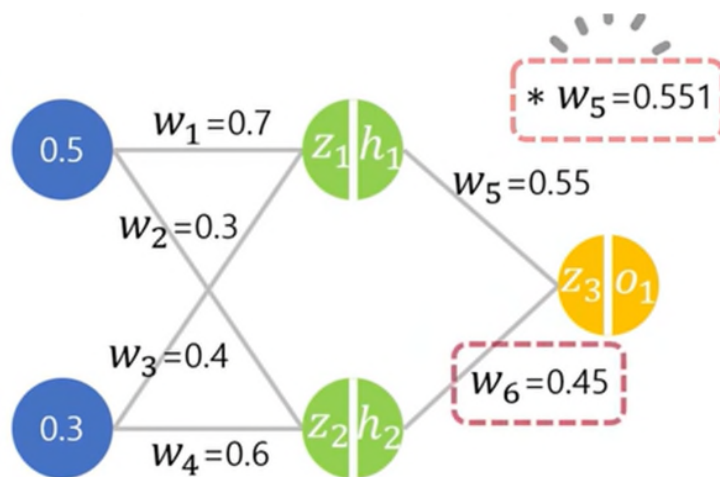
가중치 업데이트



경사하강법을 이용한 가중치 학습법:

$$0.551 = 0.55 + 0.01 \times 0.1$$

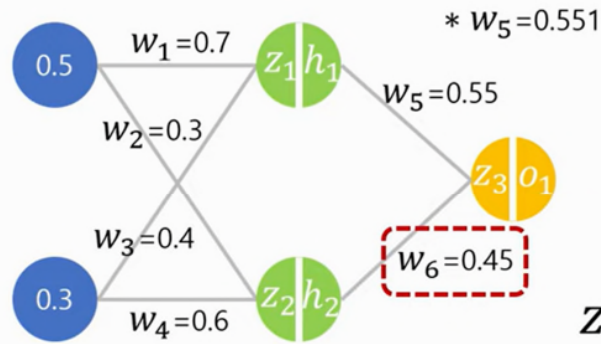
590



$$\frac{\partial C}{\partial w_6} = \frac{\partial C}{\partial o_1} \cdot \frac{\partial o_1}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_6}$$

이 부분은 앞서 계산 한 값과 동일합니다

591



$$z_3 = h_1 w_5 + h_2 w_6$$

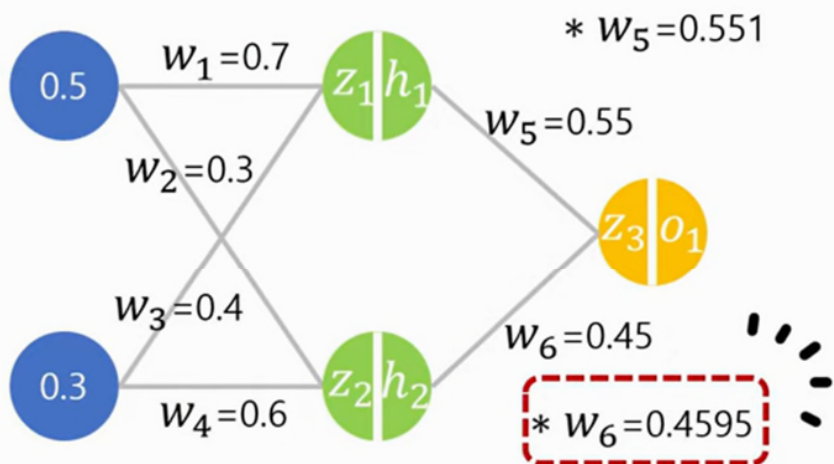
$$\frac{\partial z_3}{\partial w_6} = \cancel{h_1 w_5} + h_2 \cancel{w_6}$$

$$= h_2 = 0.582$$

$$\frac{\partial C}{\partial w_6} = -0.71 \cdot 0.229 \cdot \frac{\partial z_3}{\partial w_6}$$

$$= -0.095$$

592

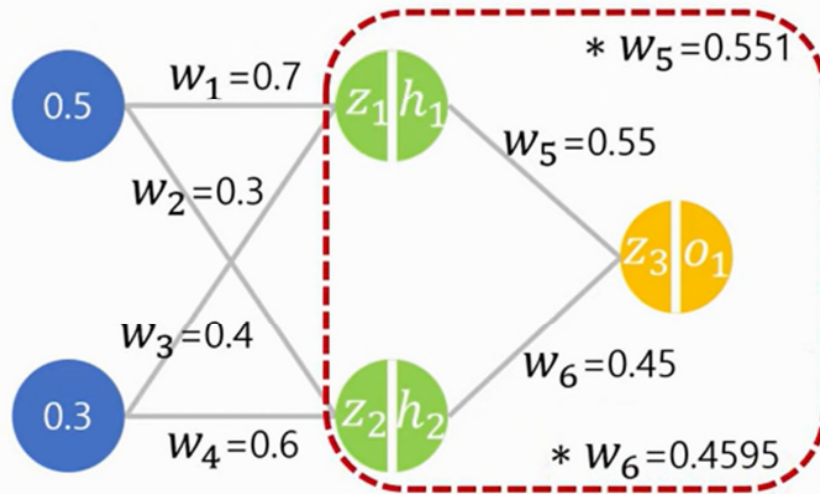


경사하강법을 이용한 가중치 학습법:

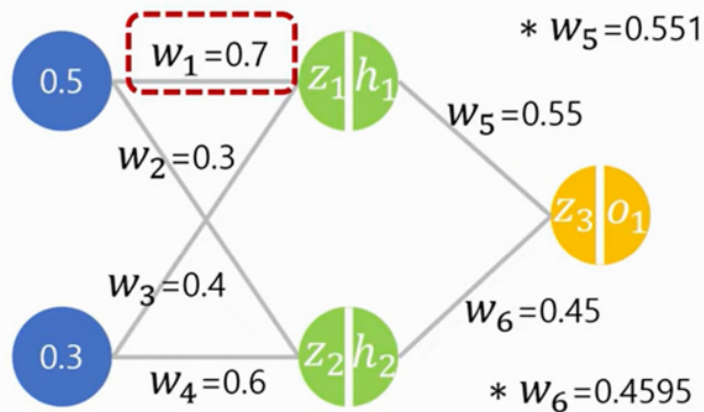
$$0.4595 = 0.71 \cdot 0.45 + 0.095 \times 0.1$$

593

2층은 가중치 업데이트가 완료되었습니다

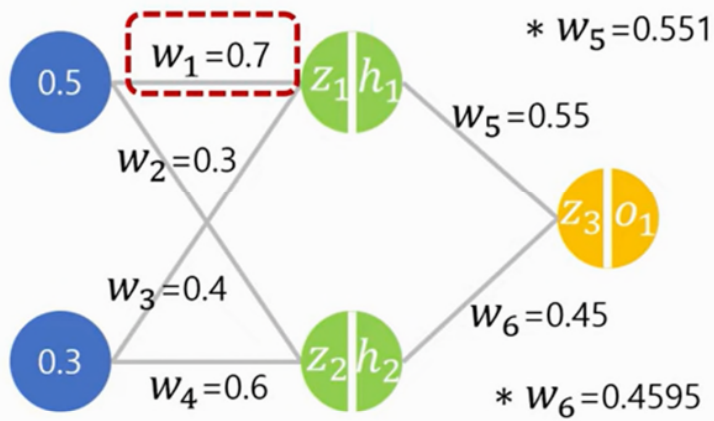


594



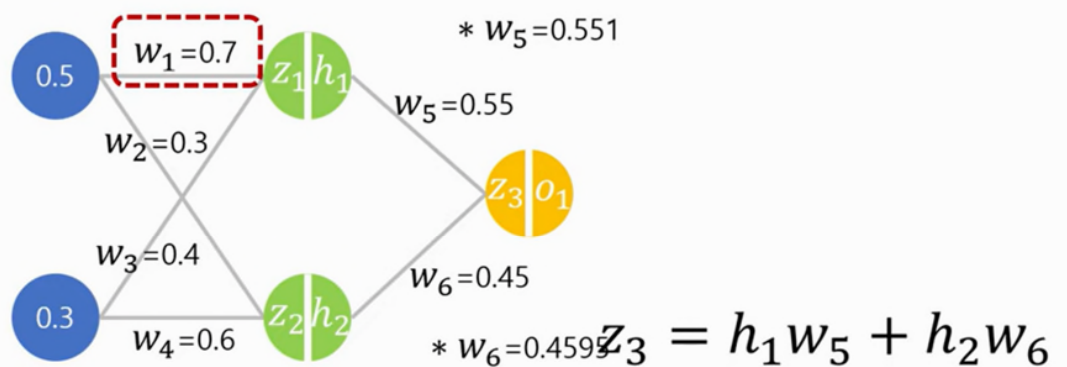
$$\frac{\partial C}{\partial w_1} = \frac{\partial C}{\partial o_1} \cdot \frac{\partial o_1}{\partial z_3} \cdot \frac{\partial z_3}{\partial h_1} \cdot \frac{\partial h_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1}$$

595



$$\frac{\partial C}{\partial w_1} = \frac{\frac{\partial C}{\partial o_1} \cdot \frac{\partial o_1}{\partial z_3}}{-0.71 \cdot 0.229} \cdot \frac{\partial z_3}{\partial h_1} \cdot \frac{\partial h_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1}$$

596

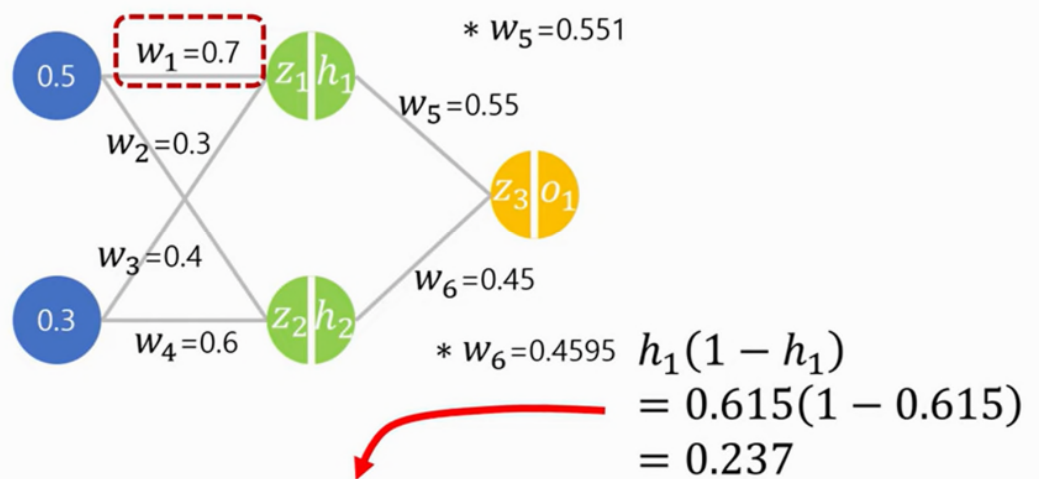


$$\frac{\partial C}{\partial w_1} = \frac{\frac{\partial C}{\partial o_1} \cdot \frac{\partial o_1}{\partial z_3}}{-0.71 \cdot 0.229} \cdot \frac{\frac{\partial z_3}{\partial h_1}}{0.55} \cdot \frac{\partial h_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1} = w_5 = 0.55$$

$\frac{\partial z_3}{\partial h_1} = \cancel{h_1w_5} + \cancel{h_2w_6} = w_5 = 0.55$



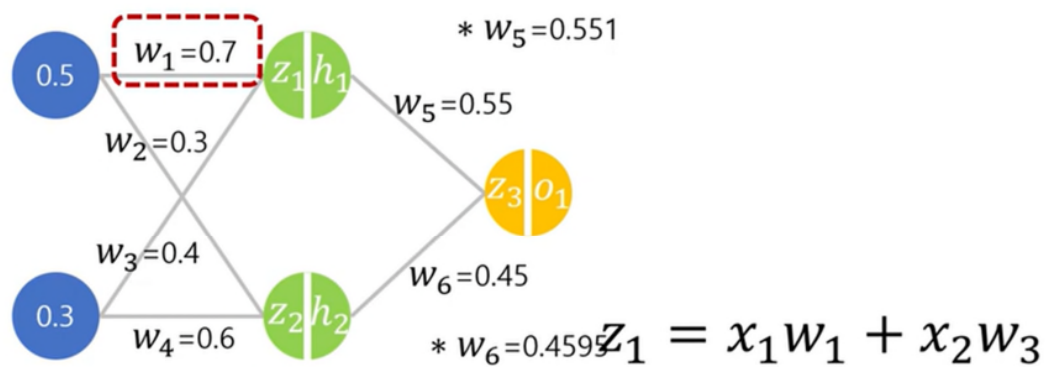
597



$$\frac{\partial C}{\partial w_1} = \frac{\partial C}{\partial o_1} \cdot \frac{\partial o_1}{\partial z_3} \cdot \frac{\partial z_3}{\partial h_1} \cdot \frac{\partial h_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1}$$

-0.71 · 0.229 0.55

598



$$\frac{\partial C}{\partial w_1} = \frac{\partial C}{\partial o_1} \cdot \frac{\partial o_1}{\partial z_3} \cdot \frac{\partial z_3}{\partial h_1} \cdot \frac{\partial h_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1}$$

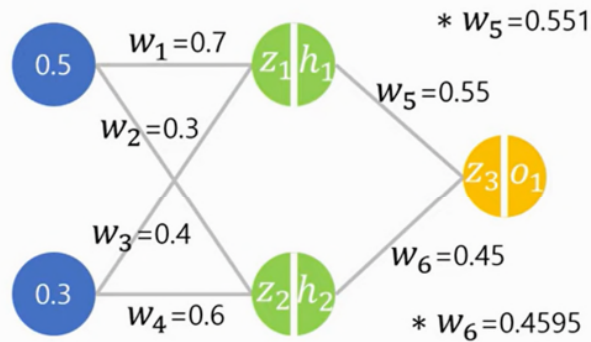
-0.71 · 0.229 0.55 0.237 0.5

$$\frac{\partial z_1}{\partial w_1} = x_1 \cancel{w_1} + \cancel{x_2 w_3} = x_1$$



599

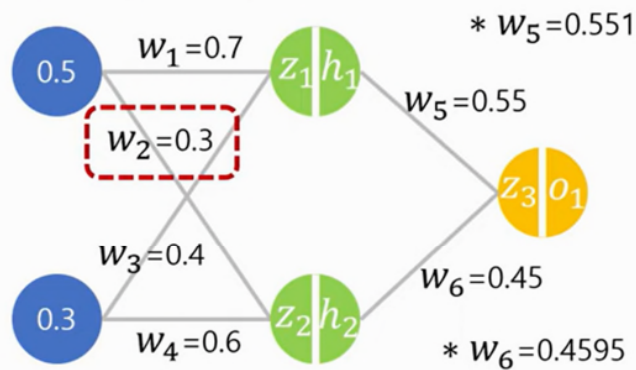
* $w_1=0.7010$



$$\frac{\partial C}{\partial w_1} = -0.0106 \quad \text{새 연결강도} = 0.7 + 0.0106 \times 0.1 = 0.7010$$

600

* $w_1=0.7010$

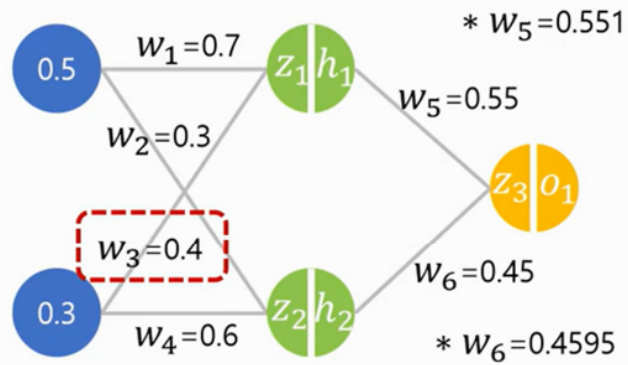


$$\frac{\partial C}{\partial w_2} = \frac{\partial C}{\partial o_1} \cdot \frac{\partial o_1}{\partial z_3} \cdot \frac{\partial z_3}{\partial h_2} \cdot \frac{\partial h_2}{\partial z_2} \cdot \frac{\partial z_2}{\partial w_2} = -0.009$$

$-0.71 \cdot 0.229$ $\frac{w_6}{=0.45}$ $\frac{h_2(1-h_2)}{=0.243}$ $\frac{x_1}{=0.5}$

601

* $w_1=0.7010$

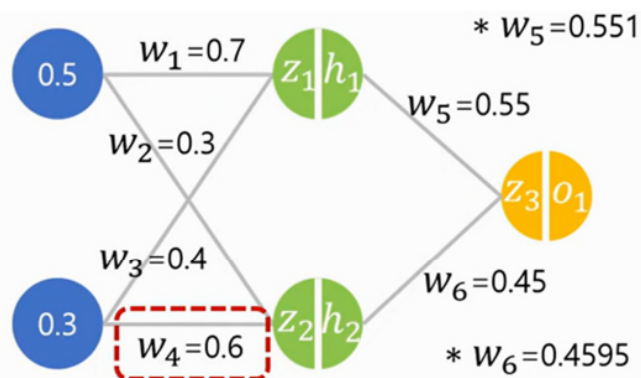


$$\frac{\partial C}{\partial w_3} = \frac{\partial C}{\partial o_1} \cdot \frac{\partial o_1}{\partial z_3} \cdot \frac{\partial z_3}{\partial h_1} \cdot \frac{\partial h_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_3} = -0.011$$

$-0.71 \cdot 0.229$ $\frac{w_5}{=0.55}$ $\frac{h_1(1-h_1)}{=0.237}$ $\frac{x_2}{=0.5}$

602

* $w_1=0.7010$



$$\frac{\partial C}{\partial w_4} = \frac{\partial C}{\partial o_1} \cdot \frac{\partial o_1}{\partial z_3} \cdot \frac{\partial z_3}{\partial h_2} \cdot \frac{\partial h_2}{\partial z_2} \cdot \frac{\partial z_2}{\partial w_4} = -0.005$$

$-0.71 \cdot 0.229$ $\frac{w_6}{=0.45}$ $\frac{h_2(1-h_2)}{=0.243}$ $\frac{x_1}{=0.3}$

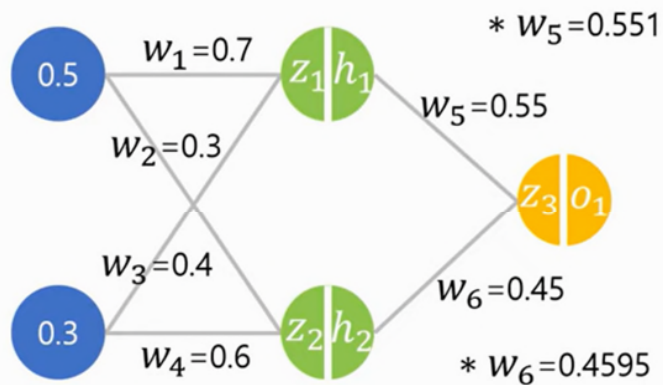
603

$$* w_1=0.7010$$

$$* w_2=0.3009$$

$$* w_3=0.4011$$

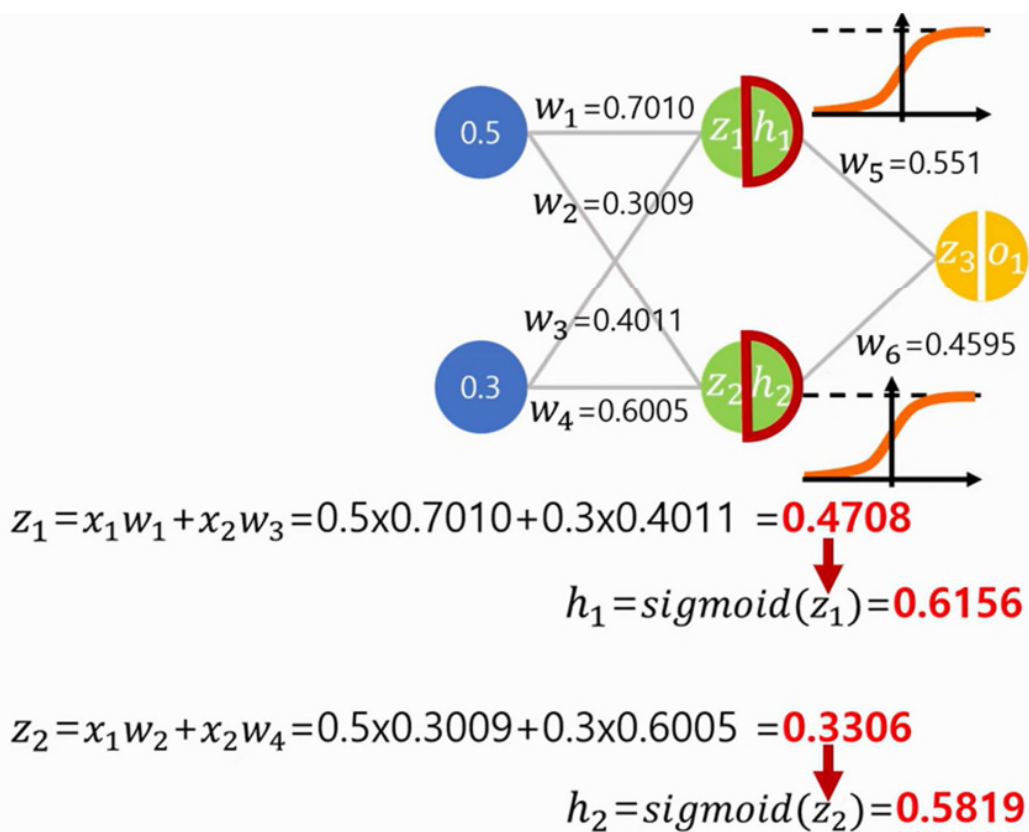
$$* w_4=0.6005$$



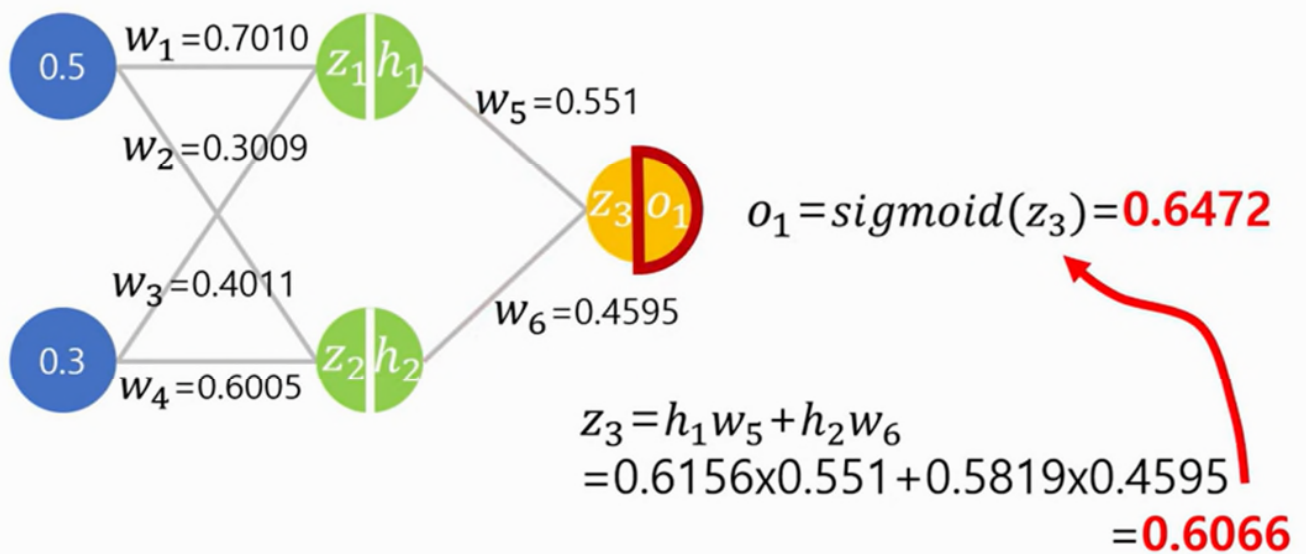
$$\frac{\partial C}{\partial w_2} = -0.009 \quad \frac{\partial C}{\partial w_3} = -0.011 \quad \frac{\partial C}{\partial w_4} = -0.005$$

604

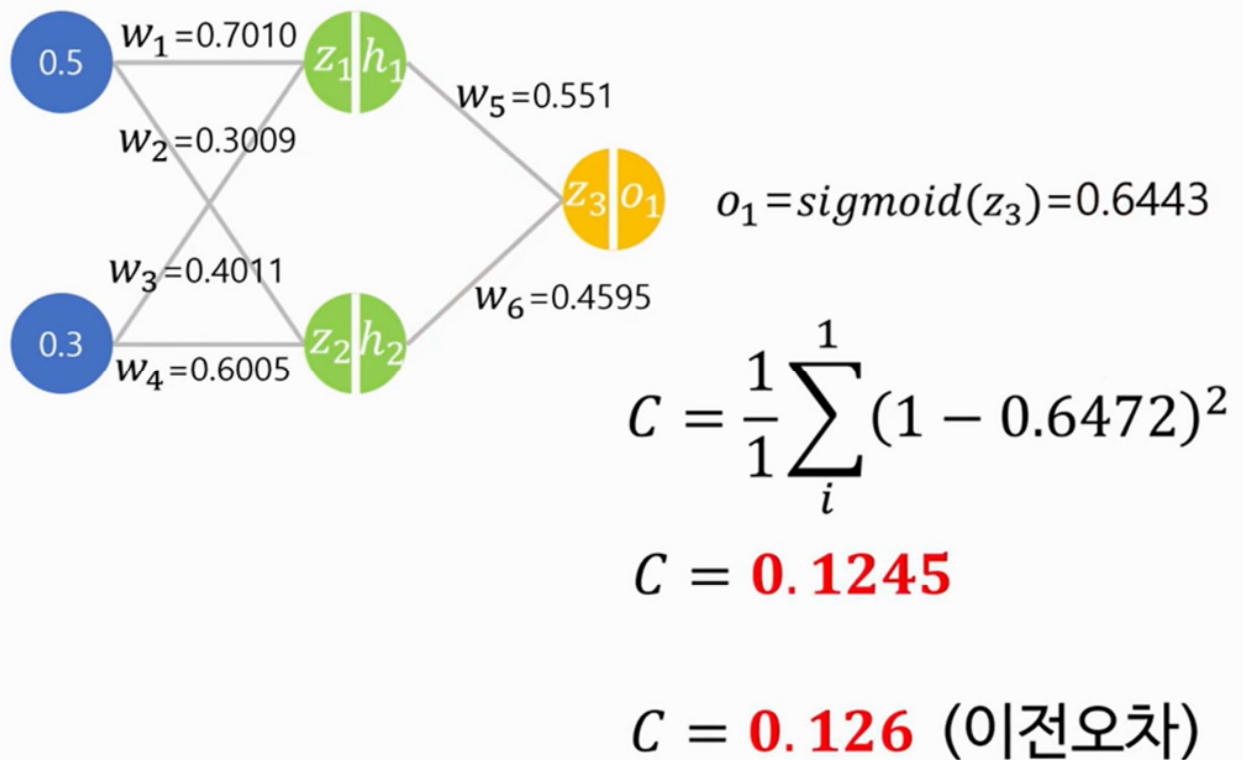
새로 업데이트한 파라미터를 사용하여
순전파 진행 -> Loss가 줄어드렸는지 계산



605

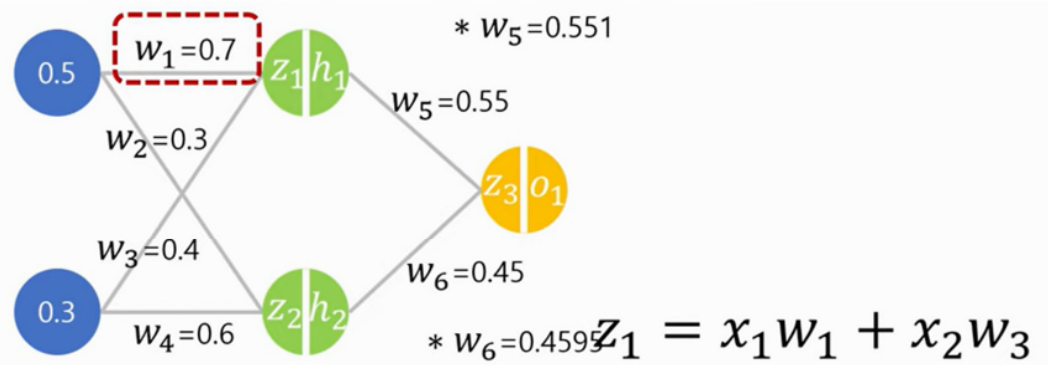


606



607

반복하여 역전파 실행하여 파라미터 업데이트 실행



$$\frac{\partial C}{\partial w_1} = \frac{\partial C}{\partial o_1} \cdot \frac{\partial o_1}{\partial z_3} \cdot \frac{\partial z_3}{\partial h_1} \cdot \frac{\partial h_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1}$$

Below the equation, the values for each term are listed: $-0.71 \cdot 0.229 \cdot 0.55 \cdot 0.237 \cdot 0.5$.

Below the diagram, the equation $z_1 = x_1 w_1 + x_2 w_3$ is shown. The derivative $\frac{\partial z_1}{\partial w_1} = x_1 w_1 + x_2 w_3$ is shown with the second term crossed out, resulting in $= x_1$. A red arrow points from this result to the $\frac{\partial z_1}{\partial w_1}$ term in the chain rule equation above.

